

FALL Semester 2026-027
BITE305P - Computer Networks
Assessment 1

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1. Study and test the functionality of basic networking commands in the laboratory.

A.) PING

i) What is the IP address of www.vit.ac.in ?

```
pushp@PushpGoel:/mnt/c/Users/pushp$ ping www.vit.ac.in
PING vit.ac.in (122.184.65.22) 56(84) bytes of data.
```

ii) Check whether TCP/IP is properly installed and functioning in your system.

```
pushp@PushpGoel:/mnt/c/Users/pushp$ ping 127.0.0.1
PING 127.0.0.1 (127.0.0.1) 56(84) bytes of data.
64 bytes from 127.0.0.1: icmp_seq=1 ttl=64 time=2.58 ms
64 bytes from 127.0.0.1: icmp_seq=2 ttl=64 time=0.079 ms
64 bytes from 127.0.0.1: icmp_seq=3 ttl=64 time=0.073 ms
64 bytes from 127.0.0.1: icmp_seq=4 ttl=64 time=0.077 ms
64 bytes from 127.0.0.1: icmp_seq=5 ttl=64 time=0.074 ms
64 bytes from 127.0.0.1: icmp_seq=6 ttl=64 time=0.087 ms
64 bytes from 127.0.0.1: icmp_seq=7 ttl=64 time=0.079 ms
64 bytes from 127.0.0.1: icmp_seq=8 ttl=64 time=0.093 ms
64 bytes from 127.0.0.1: icmp_seq=9 ttl=64 time=0.057 ms
64 bytes from 127.0.0.1: icmp_seq=10 ttl=64 time=0.038 ms
64 bytes from 127.0.0.1: icmp_seq=11 ttl=64 time=0.061 ms
64 bytes from 127.0.0.1: icmp_seq=12 ttl=64 time=0.075 ms
64 bytes from 127.0.0.1: icmp_seq=13 ttl=64 time=0.119 ms
64 bytes from 127.0.0.1: icmp_seq=14 ttl=64 time=0.035 ms
64 bytes from 127.0.0.1: icmp_seq=15 ttl=64 time=0.057 ms
64 bytes from 127.0.0.1: icmp_seq=16 ttl=64 time=0.049 ms
64 bytes from 127.0.0.1: icmp_seq=17 ttl=64 time=0.074 ms
64 bytes from 127.0.0.1: icmp_seq=18 ttl=64 time=0.074 ms
64 bytes from 127.0.0.1: icmp_seq=19 ttl=64 time=0.075 ms
64 bytes from 127.0.0.1: icmp_seq=20 ttl=64 time=0.065 ms
64 bytes from 127.0.0.1: icmp_seq=21 ttl=64 time=0.044 ms
```

- iii) Indicate what percentage of packets sent resulted in a successful response. For the packets from which you received a response, write down the minimum, average and maximum round trip times in milliseconds.

```
pushp@PushpGoel:/mnt/c/Users/pushp$ ping -c 5 google.com
PING google.com (142.251.220.110) 56(84) bytes of data.
64 bytes from pnmaaa-be-in-f14.1e100.net (142.251.220.110): icmp_seq=1 ttl=115 time=8.84 ms
64 bytes from pnmaaa-be-in-f14.1e100.net (142.251.220.110): icmp_seq=2 ttl=115 time=9.35 ms
64 bytes from pnmaaa-be-in-f14.1e100.net (142.251.220.110): icmp_seq=3 ttl=115 time=9.71 ms
64 bytes from pnmaaa-be-in-f14.1e100.net (142.251.220.110): icmp_seq=4 ttl=115 time=8.98 ms
64 bytes from pnmaaa-be-in-f14.1e100.net (142.251.220.110): icmp_seq=5 ttl=115 time=9.07 ms

--- google.com ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4007ms
rtt min/avg/max/mdev = 8.835/9.187/9.706/0.308 ms
```

- iv) Increase or Decrease the Time Interval between Packets. Write the response.

```
pushp@PushpGoel:/mnt/c/Users/pushp$ ping -i 2 google.com
PING google.com (142.251.220.110) 56(84) bytes of data.
64 bytes from hkg07s52-in-f14.1e100.net (142.251.220.110): icmp_seq=1 ttl=115 time=8.87 ms
64 bytes from hkg07s52-in-f14.1e100.net (142.251.220.110): icmp_seq=2 ttl=115 time=9.55 ms
64 bytes from hkg07s52-in-f14.1e100.net (142.251.220.110): icmp_seq=3 ttl=115 time=8.49 ms
64 bytes from hkg07s52-in-f14.1e100.net (142.251.220.110): icmp_seq=4 ttl=115 time=8.61 ms
64 bytes from hkg07s52-in-f14.1e100.net (142.251.220.110): icmp_seq=5 ttl=115 time=9.40 ms
64 bytes from hkg07s52-in-f14.1e100.net (142.251.220.110): icmp_seq=6 ttl=115 time=8.09 ms
64 bytes from hkg07s52-in-f14.1e100.net (142.251.220.110): icmp_seq=7 ttl=115 time=9.14 ms
64 bytes from hkg07s52-in-f14.1e100.net (142.251.220.110): icmp_seq=8 ttl=115 time=9.25 ms
^C
--- google.com ping statistics ---
8 packets transmitted, 8 received, 0% packet loss, time 14019ms
rtt min/avg/max/mdev = 8.094/8.925/9.550/0.467 ms
```

- v) Send 6 packets. Indicate what percentage of the packets resulted in a successful response.

```
pushp@PushpGoel:/mnt/c/Users/pushp$ ping -c 6 google.com
PING google.com (142.251.220.110) 56(84) bytes of data.
64 bytes from pnmaaa-be-in-f14.1e100.net (142.251.220.110): icmp_seq=1 ttl=115 time=8.40 ms
64 bytes from pnmaaa-be-in-f14.1e100.net (142.251.220.110): icmp_seq=2 ttl=115 time=8.75 ms
64 bytes from pnmaaa-be-in-f14.1e100.net (142.251.220.110): icmp_seq=3 ttl=115 time=9.63 ms
64 bytes from pnmaaa-be-in-f14.1e100.net (142.251.220.110): icmp_seq=4 ttl=115 time=8.68 ms
64 bytes from pnmaaa-be-in-f14.1e100.net (142.251.220.110): icmp_seq=5 ttl=115 time=9.45 ms
64 bytes from pnmaaa-be-in-f14.1e100.net (142.251.220.110): icmp_seq=6 ttl=115 time=10.2 ms

--- google.com ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 5009ms
rtt min/avg/max/mdev = 8.403/9.190/10.232/0.634 ms
```

- vi) Print Only Ping Command Summary Statistics.

```
pushp@PushpGoel:/mnt/c/Users/pushp$ ping -c 5 google.com
PING google.com (142.251.220.110) 56(84) bytes of data.
64 bytes from hkg07s52-in-f14.1e100.net (142.251.220.110): icmp_seq=1 ttl=115 time=9.03 ms
64 bytes from hkg07s52-in-f14.1e100.net (142.251.220.110): icmp_seq=2 ttl=115 time=10.2 ms
64 bytes from hkg07s52-in-f14.1e100.net (142.251.220.110): icmp_seq=3 ttl=115 time=8.92 ms
64 bytes from hkg07s52-in-f14.1e100.net (142.251.220.110): icmp_seq=4 ttl=115 time=9.06 ms
64 bytes from hkg07s52-in-f14.1e100.net (142.251.220.110): icmp_seq=5 ttl=115 time=9.27 ms

--- google.com ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4006ms
rtt min/avg/max/mdev = 8.916/9.305/10.249/0.485 ms
```

vii) Change Ping Packet Size of 512, 1024 bytes of data. Write the response.

```
pushp@PushpGoel:/mnt/c/Users/pushp$ ping -s 512 google.com -c 4
PING google.com (142.251.220.110) 512(540) bytes of data.
520 bytes from pnmaaa-be-in-f14.1e100.net (142.251.220.110): icmp_seq=1 ttl=115 time=9.58 ms
520 bytes from pnmaaa-be-in-f14.1e100.net (142.251.220.110): icmp_seq=2 ttl=115 time=8.87 ms
520 bytes from pnmaaa-be-in-f14.1e100.net (142.251.220.110): icmp_seq=3 ttl=115 time=9.16 ms
520 bytes from pnmaaa-be-in-f14.1e100.net (142.251.220.110): icmp_seq=4 ttl=115 time=9.22 ms

--- google.com ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3004ms
rtt min/avg/max/mdev = 8.867/9.206/9.576/0.252 ms
pushp@PushpGoel:/mnt/c/Users/pushp$ ping -s 1024 google.com -c 4
PING google.com (142.251.220.110) 1024(1052) bytes of data.
1032 bytes from hkg07s52-in-f14.1e100.net (142.251.220.110): icmp_seq=1 ttl=115 time=9.13 ms
1032 bytes from hkg07s52-in-f14.1e100.net (142.251.220.110): icmp_seq=2 ttl=115 time=9.61 ms
1032 bytes from hkg07s52-in-f14.1e100.net (142.251.220.110): icmp_seq=3 ttl=115 time=9.04 ms
1032 bytes from hkg07s52-in-f14.1e100.net (142.251.220.110): icmp_seq=4 ttl=115 time=8.37 ms

--- google.com ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3005ms
rtt min/avg/max/mdev = 8.367/9.035/9.605/0.441 ms
```

viii) Execute the Timeout option for 5 seconds.

```
pushp@PushpGoel:/mnt/c/Users/pushp$ ping -W 5 google.com -c 4
PING google.com (142.251.220.110) 56(84) bytes of data.
64 bytes from pnmaaa-be-in-f14.1e100.net (142.251.220.110): icmp_seq=1 ttl=115 time=8.92 ms
64 bytes from pnmaaa-be-in-f14.1e100.net (142.251.220.110): icmp_seq=2 ttl=115 time=9.40 ms
64 bytes from pnmaaa-be-in-f14.1e100.net (142.251.220.110): icmp_seq=3 ttl=115 time=8.94 ms
64 bytes from pnmaaa-be-in-f14.1e100.net (142.251.220.110): icmp_seq=4 ttl=115 time=9.32 ms

--- google.com ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3006ms
rtt min/avg/max/mdev = 8.921/9.146/9.398/0.215 ms
```

ix) Write the current version of ping program.

```
pushp@PushpGoel:/mnt/c/Users/pushp$ ping -V
ping from iputils 20250605
libcap: yes, IDN: yes, NLS: no, error.h: yes, getrandom(): yes, __fpending(): yes
```

x) For some of the hosts, you may not have received any responses for the packets you have sent. What are the reasons for not getting a response?

Some hosts may not respond to ping requests because they block ICMP packets through a firewall, the host may be offline, the IP address or hostname may be incorrect, there may be network or routing issues, or the server is configured to ignore ping requests for security reasons.

B.) IFCONFIG

i) What is the MAC address of the network interface card of your system?

```
pushp@PushpGoel:/mnt/c/Users/pushp$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 172.31.114.232 netmask 255.255.240.0 broadcast 172.31.127.255
    inet6 fe80::215:5dff:fe3e:cd86 prefixlen 64 scopeid 0x20<link>
    ether 00:15:5d:3e:cd:86 txqueuelen 1000 (Ethernet)
    RX packets 13322 bytes 36275734 (36.2 MB)
```

ii) What is the IP address of your system?

```
pushp@PushpGoel:/mnt/c/Users/pushp$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 172.31.114.232 netmask 255.255.240.0 broadcast 172.31.127.255
    inet6 fe80::215:5dff:fe3e:cd86 prefixlen 64 scopeid 0x20<link>
    ether 00:15:5d:3e:cd:86 txqueuelen 1000 (Ethernet)
    RX packets 13322 bytes 36275734 (36.2 MB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 10423 bytes 733004 (733.0 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 202 bytes 18973 (18.9 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 202 bytes 18973 (18.9 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

iii) Find out all the network interfaces connected to your system.

```
pushp@PushpGoel:/mnt/c/Users/pushp$ ifconfig -a
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 172.31.114.232 netmask 255.255.240.0 broadcast 172.31.127.255
    inet6 fe80::215:5dff:fe3e:cd86 prefixlen 64 scopeid 0x20<link>
    ether 00:15:5d:3e:cd:86 txqueuelen 1000 (Ethernet)
    RX packets 13322 bytes 36275734 (36.2 MB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 10423 bytes 733004 (733.0 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 202 bytes 18973 (18.9 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 202 bytes 18973 (18.9 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

iv) Display the Network Settings of Specific Interface.

```
pushp@PushpGoel:/mnt/c/Users/pushp$ ifconfig eth0
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 172.31.114.232 netmask 255.255.240.0 broadcast 172.31.127.255
    inet6 fe80::215:5dff:fe3e:cd86 prefixlen 64 scopeid 0x20<link>
    ether 00:15:5d:3e:cd:86 txqueuelen 1000 (Ethernet)
    RX packets 13322 bytes 36275734 (36.2 MB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 10423 bytes 733004 (733.0 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

C.) HOSTNAME

i) Find the name of your system?

```
pushp@PushpGoel:/mnt/c/Users/pushp$ hostname
PushpGoel
```

ii) What is the significance of the name?

The hostname uniquely identifies a computer on a network and helps in communication and system administration.

D.) NETSTAT

i) List Various Listening Ports.

```
pushp@PushpGoel:/mnt/c/Users/pushp$ netstat -l
Active Internet connections (only servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
tcp      0      0 10.255.255.254:domain  0.0.0.0:*               LISTEN
tcp      0      0 127.0.0.53:domain      0.0.0.0:*               LISTEN
tcp      0      0 127.0.0.54:domain      0.0.0.0:*               LISTEN
udp      0      0 127.0.0.54:domain      0.0.0.0:*
udp      0      0 127.0.0.53:domain      0.0.0.0:*
udp      0      0 10.255.255.254:domain  0.0.0.0:*
udp      0      0 localhost:323           0.0.0.0:*
udp      0      0 localhost:323           0.0.0.0:*
udp6     0      0 ip6-localhost:323      [::]:*
udp6     0      0 ip6-localhost:323      [::]:*

Active UNIX domain sockets (only servers)
Proto RefCnt Flags               Type               State         I-Node  Path
unix  2      [ ACC ] STREAM           LISTENING         3115        /run/WSL/2_interop
unix  2      [ ACC ] STREAM           LISTENING         7190        /run/WSL/1_interop
unix  2      [ ACC ] SEQPACKET        LISTENING         6181        /mnt/wslg/weston-notify.sock
unix  2      [ ACC ] STREAM           LISTENING         1952        /run/dbus/system_bus_socket
unix  2      [ ACC ] STREAM           LISTENING         4100        /mnt/wslg/runtime-dir/wayland-0
unix  2      [ ACC ] STREAM           LISTENING         4103        /tmp/.X11-unix/X0
unix  2      [ ACC ] STREAM           LISTENING         5808        /mnt/wslg/runtime-dir/pulse/native
unix  2      [ ACC ] STREAM           LISTENING         5815        /mnt/wslg/PulseServer
unix  2      [ ACC ] STREAM           LISTENING         6187        /mnt/wslg/PulseAudioRDPSource
unix  2      [ ACC ] STREAM           LISTENING         690         /run/user/1000/systemd/private
unix  2      [ ACC ] STREAM           LISTENING         692         /run/user/1000/systemd/io.systemd.Manager
unix  2      [ ACC ] STREAM           LISTENING         8484        /mnt/wslg/PulseAudioRDPSink
unix  2      [ ACC ] STREAM           LISTENING         701         /run/user/1000/bus
unix  2      [ ACC ] STREAM           LISTENING         702         /run/user/1000/gnupg/S.dirmngr
unix  2      [ ACC ] STREAM           LISTENING         703         /run/user/1000/gnupg/S.gpg-agent.browser
unix  2      [ ACC ] STREAM           LISTENING         704         /run/user/1000/gnupg/S.gpg-agent.extra
unix  2      [ ACC ] STREAM           LISTENING         705         /run/user/1000/gnupg/S.gpg-agent.ssh
unix  2      [ ACC ] STREAM           LISTENING         706         /run/user/1000/gnupg/S.gpg-agent
unix  2      [ ACC ] STREAM           LISTENING         707         /run/user/1000/gnupg/S.keyboxd
unix  2      [ ACC ] STREAM           LISTENING         708         /run/user/1000/pk-debconf-socket
unix  2      [ ACC ] STREAM           LISTENING         709         /run/user/1000/snapd-session-agent.socket
unix  2      [ ACC ] STREAM           LISTENING         710         /run/user/1000/openssh_agent
unix  2      [ ACC ] STREAM           LISTENING         8715        /run/user/1000/systemd/io.systemd.AskPassword
unix  2      [ ACC ] STREAM           LISTENING         3096        /tmp/dbus-gLfq1fapmv
unix  2      [ ACC ] STREAM           LISTENING         2228        /run/systemd/private
unix  2      [ ACC ] STREAM           LISTENING         2229        /run/systemd/userdb/io.systemd.DynamicUser
unix  2      [ ACC ] STREAM           LISTENING         2230        /run/systemd/io.systemd.ManagedOOM
unix  2      [ ACC ] STREAM           LISTENING         2231        /run/systemd/io.systemd.Manager
unix  2      [ ACC ] STREAM           LISTENING         7349        /run/apport.socket
unix  2      [ ACC ] STREAM           LISTENING         2239        /run/systemd/io.systemd.AskPassword
unix  2      [ ACC ] STREAM           LISTENING         2240        /run/systemd/io.systemd.Credentials
unix  2      [ ACC ] STREAM           LISTENING         7350        /run/dbus/system_bus_socket
unix  2      [ ACC ] STREAM           LISTENING         2241        /run/systemd/io.systemd.FactoryReset
unix  2      [ ACC ] STREAM           LISTENING         2244        /run/systemd/journal/stdout
unix  2      [ ACC ] STREAM           LISTENING         7351        /run/polkit/agent-helper.socket
unix  2      [ ACC ] STREAM           LISTENING         2245        /run/systemd/io.systemd.MuteConsole
unix  2      [ ACC ] STREAM           LISTENING         2500        /run/snapd.socket
unix  2      [ ACC ] STREAM           LISTENING         2246        /run/systemd/resolve/io.systemd.Resolve.Monitor
```

ii) List TCP Ports connections

```
pushp@PushpGoel:/mnt/c/Users/pushp$ netstat -t
Active Internet connections (w/o servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
```

iii) List UDP Ports connections

```
pushp@PushpGoel:/mnt/c/Users/pushp$ netstat -u
Active Internet connections (w/o servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
```

iv) List all the LISTENING Connections

```
pushhp@PushpGoel:/mnt/c/Users/pushp$ netstat -l
Active Internet connections (only servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
tcp      0      0 10.255.255.254:domain  0.0.0.0:*              LISTEN
tcp      0      0 127.0.0.53:domain      0.0.0.0:*              LISTEN
tcp      0      0 127.0.0.54:domain      0.0.0.0:*              LISTEN
udp      0      0 127.0.0.54:domain      0.0.0.0:*
udp      0      0 127.0.0.53:domain      0.0.0.0:*
udp      0      0 10.255.255.254:domain  0.0.0.0:*
udp      0      0 localhost:323           0.0.0.0:*
udp      0      0 localhost:323           0.0.0.0:*
udp6     0      0 ip6-localhost:323      [::]:*
udp6     0      0 ip6-localhost:323      [::]:*

Active UNIX domain sockets (only servers)
Proto RefCnt Flags       Type       State       I-Node     Path
unix  2      [ ACC ] STREAM    LISTENING   3115       /run/WSL/2_interop
unix  2      [ ACC ] STREAM    LISTENING   7190       /run/WSL/1_interop
unix  2      [ ACC ] SEQPACKET LISTENING   6181       /mnt/wslg/weston-notify.sock
unix  2      [ ACC ] STREAM    LISTENING   1952       /run/dbus/system_bus_socket
unix  2      [ ACC ] STREAM    LISTENING   4100       /mnt/wslg/runtime-dir/wayland-0
unix  2      [ ACC ] STREAM    LISTENING   4103       /tmp/.X11-unix/X0
unix  2      [ ACC ] STREAM    LISTENING   5808       /mnt/wslg/runtime-dir/pulse/native
unix  2      [ ACC ] STREAM    LISTENING   5815       /mnt/wslg/PulseServer
unix  2      [ ACC ] STREAM    LISTENING   6187       /mnt/wslg/PulseAudioRDPSource
unix  2      [ ACC ] STREAM    LISTENING   690        /run/user/1000/systemd/private
unix  2      [ ACC ] STREAM    LISTENING   692        /run/user/1000/systemd/io.systemd.Manager
unix  2      [ ACC ] STREAM    LISTENING   8484       /mnt/wslg/PulseAudioRDPSink
unix  2      [ ACC ] STREAM    LISTENING   701        /run/user/1000/bus
unix  2      [ ACC ] STREAM    LISTENING   702        /run/user/1000/gnupg/S.dirmngr
unix  2      [ ACC ] STREAM    LISTENING   703        /run/user/1000/gnupg/S.gpg-agent.browser
unix  2      [ ACC ] STREAM    LISTENING   704        /run/user/1000/gnupg/S.gpg-agent.extra
unix  2      [ ACC ] STREAM    LISTENING   705        /run/user/1000/gnupg/S.gpg-agent.ssh
unix  2      [ ACC ] STREAM    LISTENING   706        /run/user/1000/gnupg/S.gpg-agent
unix  2      [ ACC ] STREAM    LISTENING   707        /run/user/1000/gnupg/S.keyboxd
unix  2      [ ACC ] STREAM    LISTENING   708        /run/user/1000/pk-debconf-socket
unix  2      [ ACC ] STREAM    LISTENING   709        /run/user/1000/snapd-session-agent.socket
```

v) Find the statistics of all protocols.

```
pushp@PushpGoe1:/mnt/c/Users/pushp$ netstat -s
Ip:
    Forwarding: 2
    11807 total packets received
    0 forwarded
    0 incoming packets discarded
    11760 incoming packets delivered
    10613 requests sent out
    OutTransmits: 10613
Icmp:
    102 ICMP messages received
    0 input ICMP message failed
    ICMP input histogram:
        echo requests: 33
        echo replies: 69
    130 ICMP messages sent
    0 ICMP messages failed
    ICMP output histogram:
        echo requests: 97
        echo replies: 33
IcmpMsg:
    InType0: 69
    InType8: 33
    OutType0: 33
    OutType8: 97
Tcp:
    16 active connection openings
    0 passive connection openings
    0 failed connection attempts
    1 connection resets received
    0 connections established
    11443 segments received
    10285 segments sent out
    0 segments retransmitted
    9 bad segments received
    7 resets sent
    InCsumErrors: 8
Udp:
    203 packets received
    0 packets to unknown port received
    0 packet receive errors
    203 packets sent
    0 receive buffer errors
    0 send buffer errors
    IgnoredMulti: 12
UdpLite:
TcpExt:
    4 TCP sockets finished time wait in fast timer
    2 delayed acks sent
    Quick ack mode was activated 1 times
    135 packet headers predicted
    22 acknowledgments not containing data payload received
```

vi) Display Kernel IP routing table.

```
pushp@PushpGoel:/mnt/c/Users/pushp$ netstat -r
Kernel IP routing table
Destination        Gateway            Genmask           Flags   MSS Window  irtt Iface
default            Pushp_Goel.msho  0.0.0.0           UG        0  0           0 eth0
172.31.112.0       0.0.0.0           255.255.240.0     U        0  0           0 eth0
```

vii) Show the Kernel interface table, similar to ifconfig command.

```
pushp@PushpGoel:/mnt/c/Users/pushp$ netstat -i
Kernel Interface table
Iface      MTU      RX-OK RX-ERR RX-DRP RX-OVR    TX-OK TX-ERR TX-DRP TX-OVR Flg
eth0       1500     13329    0      0  0      10431    0      0      0 BMRU
lo         65536     216      0      0  0       216      0      0      0 LRU
```

viii) By simply opening a browser connection to HTTP (port 80) server (while still offline!) what will be status of *netstat* command?

```
pushp@PushpGoel:/mnt/c/Users/pushp$ netstat -ant
Active Internet connections (servers and established)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
tcp    0      0 10.255.255.254:53       0.0.0.0:*               LISTEN
tcp    0      0 127.0.0.53:53          0.0.0.0:*               LISTEN
tcp    0      0 127.0.0.54:53          0.0.0.0:*               LISTEN
```

E.) TRACEROUTE or TRACERT

i) Write the function of this command.

```
pushp@PushpGoel:/mnt/c/Users/pushp$ traceroute google.com
traceroute to google.com (142.251.220.110), 30 hops max, 60 byte packets
 1 Pushp_Goel.mshome.net (172.31.112.1)  0.387 ms  0.326 ms  0.315 ms
 2 172.16.24.1 (172.16.24.1)  4.877 ms  4.865 ms  4.855 ms
 3 abts-north-dynamic-99.218.66.182.airtelbroadband.in (182.66.218.99)  4.760 ms  6.585 ms  6.577 ms
 4 182.79.14.89 (182.79.14.89)  6.289 ms  6.237 ms  6.228 ms
 5 * * *
 6 182.79.198.0 (182.79.198.0)  9.453 ms  116.119.161.32 (116.119.161.32)  8.219 ms  182.79.142.222 (182.79.142.222)  8.201 ms
 7 * * *
 8 142.251.71.193 (142.251.71.193)  9.450 ms  142.251.70.211 (142.251.70.211)  9.394 ms  142.251.71.193 (142.251.71.193)  8.795 ms
 9 142.251.55.221 (142.251.55.221)  8.795 ms  8.779 ms  8.766 ms
10 hkg07s52-in-f14.1e100.net (142.251.220.110)  8.769 ms  8.755 ms  8.739 ms
```

F.) NSLOOKUP

i) What is the IP address and name of the machine www.google.com?

```
pushp@PushpGoel:/mnt/c/Users/pushp$ nslookup www.google.com
Server:      10.255.255.254
Address:     10.255.255.254#53

Non-authoritative answer:
Name:   www.google.com
Address: 142.251.151.119
Name:   www.google.com
Address: 142.251.152.119
Name:   www.google.com
Address: 142.251.153.119
Name:   www.google.com
Address: 142.251.154.119
Name:   www.google.com
Address: 142.251.155.119
Name:   www.google.com
Address: 142.251.156.119
Name:   www.google.com
Address: 142.251.157.119
Name:   www.google.com
Address: 142.251.150.119
Name:   www.google.com
Address: 2001:4860:482a:7700::
Name:   www.google.com
Address: 2001:4860:482b:7700::
Name:   www.google.com
Address: 2001:4860:482c:7700::
Name:   www.google.com
Address: 2001:4860:482d:7700::
Name:   www.google.com
Address: 2001:4860:4826:7700::
Name:   www.google.com
Address: 2001:4860:4827:7700::
Name:   www.google.com
Address: 2001:4860:4828:7700::
Name:   www.google.com
Address: 2001:4860:4829:7700::
```

G.) ARP

```
pushp@PushpGoel:/mnt/c/Users/pushp$ arp -a  
Pushp_Goel.mshome.net (172.31.112.1) at 00:15:5d:2f:aa:73 [ether] on eth0
```